BATPING (Bidirectional Acoustics Traveltime Pulser for INdex Gradients) 2026

Overview

Assessments from the first cycle of LSST data indicate a need to further address image degradation. In particular, while optomechanical factors are being accounted for with high levels of precision, dome and mirror seeing remains uncharacterized (Urbach et al. 2026). More specifically, temperature fluctuations within the telescope enclosure have the potential to perturb the optical path, decreasing image quality in an effect known as dome seeing. Because the index of refraction of air is temperature-dependent, turbulent pockets of warmer and colder air alike cause refraction of incoming wavefronts. This translates to a broadening of the telescope's point spread function (PSF), the size of the image in the focal plane that captures an ideal point source, like a star.

BATPING will allow us to characterize temperature fluctuations within the telescope enclosure that perturb the optical path, decreasing image quality. BATPING offers the advantage that the speed of sound is much more sensitive to variations in temperature and hence refractive index. Additionally, BATPING's ultrasonic signals, even along the optical path, do not interfere with normal observations, nor do they rely on dome illumination.

My work documented on this page will track two subprojects: (I) construction of BATPING's audio hardware components, and (II) development of a machine learning (ML) approach to optimization of the acoustic signal.

Hardware Setup

The project apparatus can be characterized as an ultrasonic thermometer--measuring environmental temperature based on deviations in the speed of an ultrasonic pulse across a set path. Such a pulse is supplied by piezeoelectric ultrasonic transducers, organized into a hexagonal array to optimize amplitude and directionality (via near-field destructive interference); the pulse is reflected and then measured by an ultrasonic mic, such as the Dodotronic ultramic_384k_evo. Cross-correlation between the original pulse and the reflected pulse reveals the time lag.

Project Log

Entry Date	Description of Work	Progress (assessment of failure modes, future improvements)

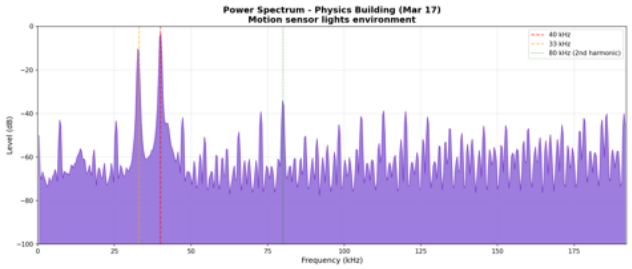
Week of 03/14/2026	• I was able to bring the sonic thermometer apparatus home with me for break, and I ran a couple of preliminary diagnostics on the hardware in preparation of reflection testing in the lab. I was able to confirm that the microphone can pick up a nicely powerful sine wave signal from the piezoelectric transducer array when it is powered by a waveform generator with a frequency of 40kHz and an amplitude of 1 V peak-to-peak. • I found that the motion-sensing light systems found in some lab settings (potentially McKay 105...) emit a standing EM pulse of ~34kHz that has the potential to saturate the Dodotronic ultramic's reception of the piezo transudcer signal. Plots showing the power spectrum in a lab setting (with motion sensors) versus at my home (without motion sensors) are shown below.	• The Dodotronic ultramic (specs) has a 384 kHz sampling rate (with 190kHz bandwidth), so I attempted to configure the Audacity sampling rate accordingly. Initially ran into issue with over-saturation of signal on 384kHz sampling rate, but this was due to the fact that the microphone has *manually adjusted* gain via a lever on the side of the mic. The issue was quickly corrected by setting the gain to switch 1 (-33dB). • When initially testing signal conservation in a lab environment, I found that the background ultrasonic pulse generated by motion-sensing light systems interfered with the microphone's ability to register the signal. However, when I tested at home in the absence of the structured noise, the microphone performed well in reading the directly transmitted signal *and* a reflected signal bounced off a right angle mirror. ☐ I'm wondering what a good metric is to measure the strength of the signal picked up by the mic relative to the original signal based on the information gathered in Audacity (which seems to record the waveform on a normalized unitless vertical axis).

L-Path Signal Reading and Structured Environmental Noise

To first confirm that the transducer apparatus was working, I tested (under two different environments) whether the microphone could pick up a transmitted signal when placed: (1)

directly in front of the piezo array (path length: 1m) and (2) aligned with a makeshift 90° reflecting mirror (path length: 2m).

Measurements in the physics building were hindered by the presence of ultrasonic motion sensors:



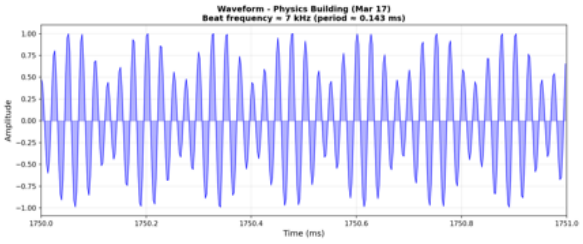


Figure 2: waveform of total received signal displaying ~7kHz beat frequency

A less noisy environment allowed for clean measurement of direct, and, as shown below, even reflected signals:

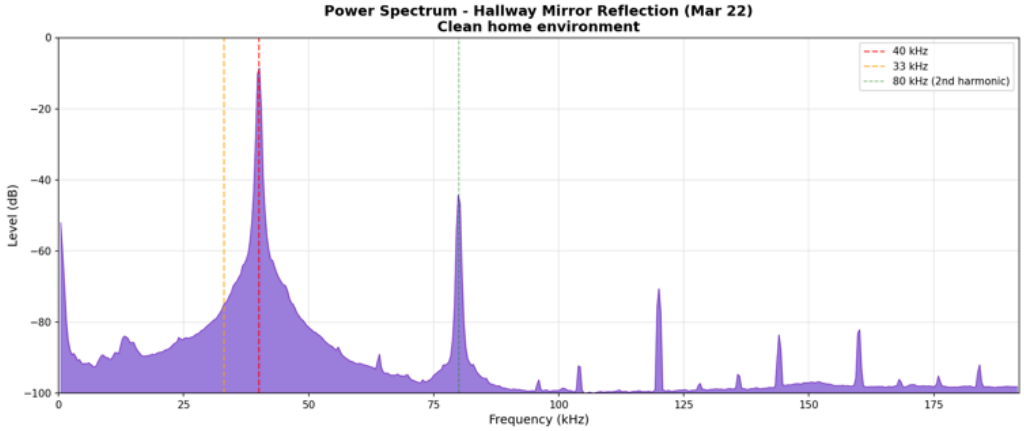
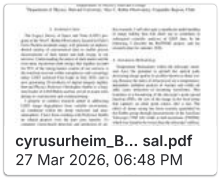


Figure 3: power spectrum recovered from L-shaped reflected path in quiet environment

Note: the vertical axis of the power spectra plots (“Level”) is measured in deciBels relative to Full Scale (dBFS), where 0 is the maximum possible amplitude the mic can pick up before clipping.

Files

Ultramic384_EVO_User_Guide.pdf



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